

BMS_SLAVE Fabrication Document

Layer Stack Legend

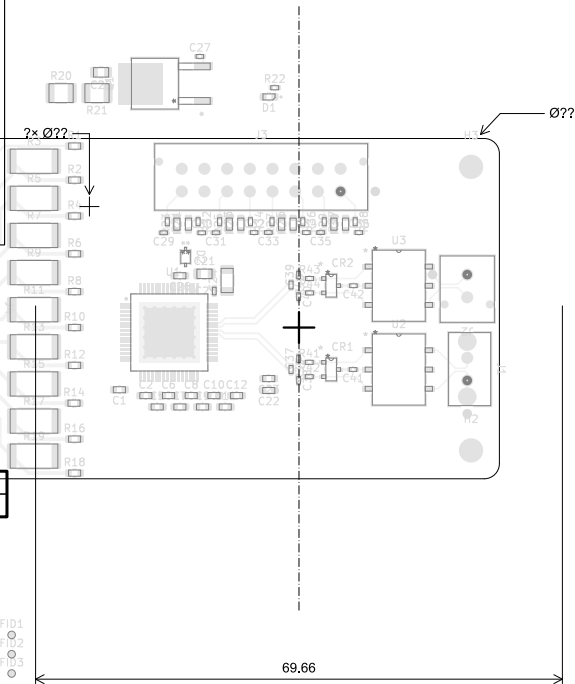
Material	Layer	Thickness	Dielectric	Type	Gerber
	F,Paste			Paste Mask	
	F,Silkscreen			Legend	GBR
	F,Mask	0,02mm		Solder Mask	GBR
Copper	L1 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	L2 (GND)	0,035mm (1,00oz)		Signal	GBR
Core		1,209994mm	FR4	Dielectric	
Copper	L3 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
Prepreg		0,1mm	FR4_7628	Dielectric	
Copper	L4 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
	B,Mask	0,02mm		Solder Mask	GBR
	B,Silkscreen			Legend	GBR
	B,Paste			Paste Mask	

Total thickness: 1,659994mm
Note: external layer thicknesses are specified after plating

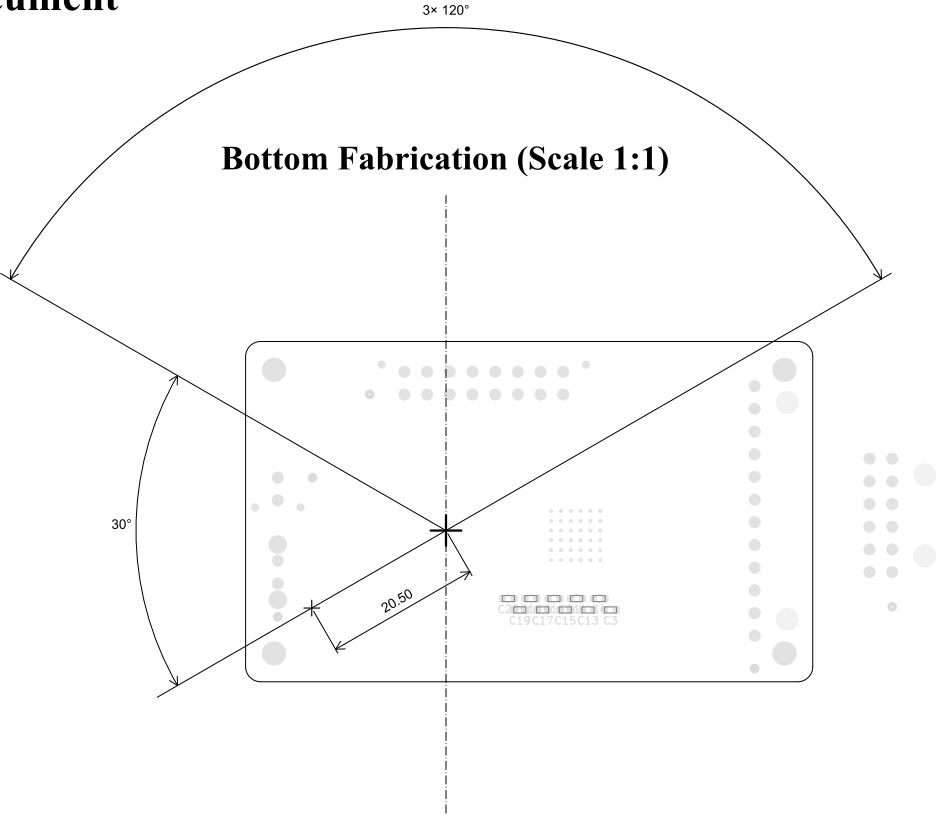
Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0,2032	0,28	L2

Top Fabrication (Scale 1:1)



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All dimensions are in millimeters unless otherwise specified.

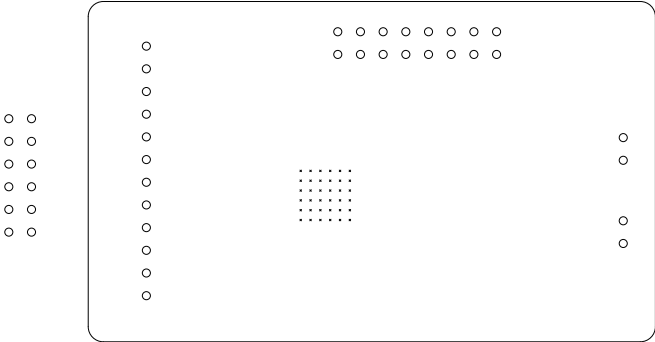
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			Blue Devil Motor Sports		PRELIMINARY		a06d4b7		
			Board Name:			Project Name:			
			BMS_SLAVE			BMS Slave			
Sheet Title:		File Name:		Designer:		Date:		Revision:	
Bottom Fabrication (Scale 1:1)		bms_slave.kicad_pcb		Aidan Brzezinski		2024-04-13		+ (Unreleased)	
Sheet Path:				Reviewer:		Size:		Sheet:	
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	36	0.20mm (8,00mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
O	44	1.07mm (42,00mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
	Total 80					

Drill Drawing L1 - L4 (Scale 1:1)





blue devil
MOTORSPORTS

Comments:

Sheet Title:
Drill Drawing (L1 - L4)

Sheet Path:

Company:
Blue Devil Motor Sports

Board Name:
BMS_SLAVE

File Name:
bms_slave.kicad_pcb

Designer:
Aidan Brzezinski

Reviewer:

Variant:
PRELIMINARY

Project Name:
BMS Slave

Date:
2024-04-13

Size:
A4

Git Hash:
a06d4b7

Revision:
+ (Unreleased)

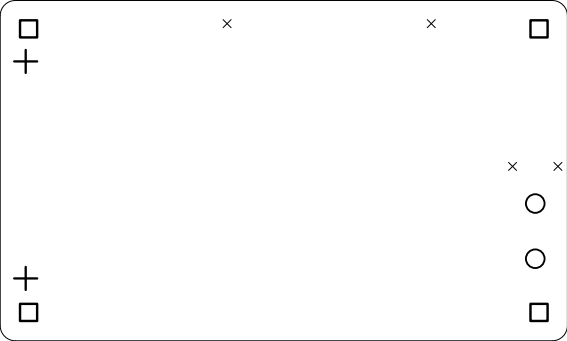
Sheet:
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	1.07mm (42.00mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
○	2	2.46mm (97.00mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
+	4	3.05mm (120.00mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
□	4	3.20mm (125.98mils)	NPTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
Total 14						

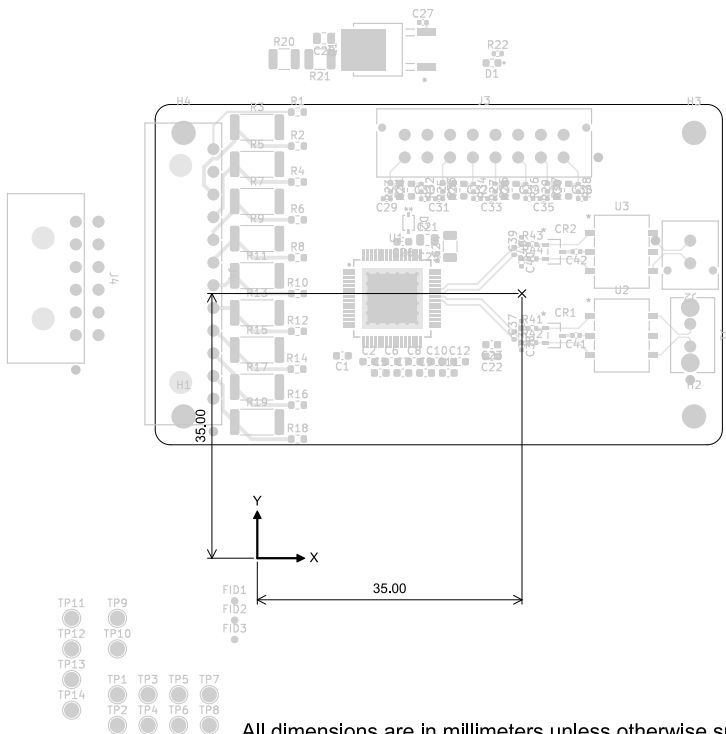
Drill Drawing L1 - L4 (Scale 1:1)



	Comments:		Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: a06d4b7
			Board Name: BMS_SLAVE		Project Name: BMS Slave	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)	
	Sheet Path:		Reviewer:	Size: A4	Sheet: 4 of 10	

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Top Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
TP1	CVDD	-5.00	-33.05
TP2	DVDD	-5.00	-37.10
TP3	AVDD	-0.95	-33.05
TP4	LDOIN	-0.95	-37.10
TP5	VCOLL	3.10	-33.05
TP6	LDOIN	3.10	-37.10
TP7	TSREF	7.15	-33.05
TP8	GND	7.15	-37.10
TP9	T_P1	-5.00	-22.95
TP10	GND	-5.00	-27.00
TP11	U1_COMLP	-11.05	-22.95
TP12	U1_COMLN	-11.05	-27.00
TP13	U1_COMHP	-11.05	-31.05
TP14	U1_COMHN	-11.05	-35.10

Ref.	Net	X [mm]	Y [mm]
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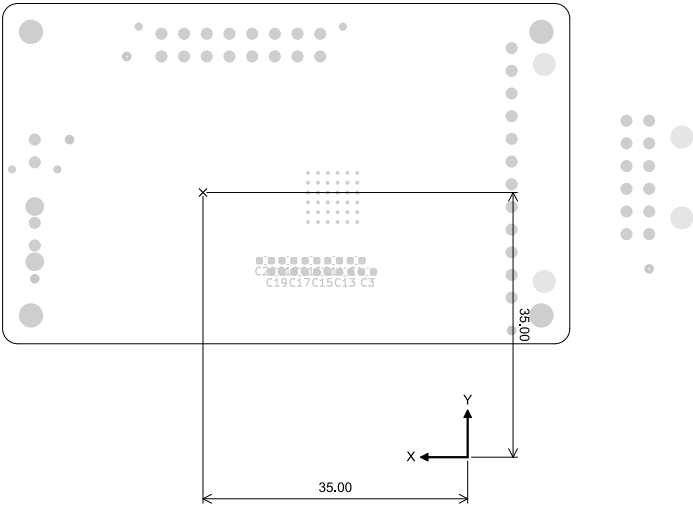
All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: a06d4b7
	Sheet Title: Top Test Points (Scale 1:1)	Board Name: BMS_SLAVE		Project Name: BMS Slave	
	Sheet Path:	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
		Reviewer:		Size: A4	Sheet: 5 of 10

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Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
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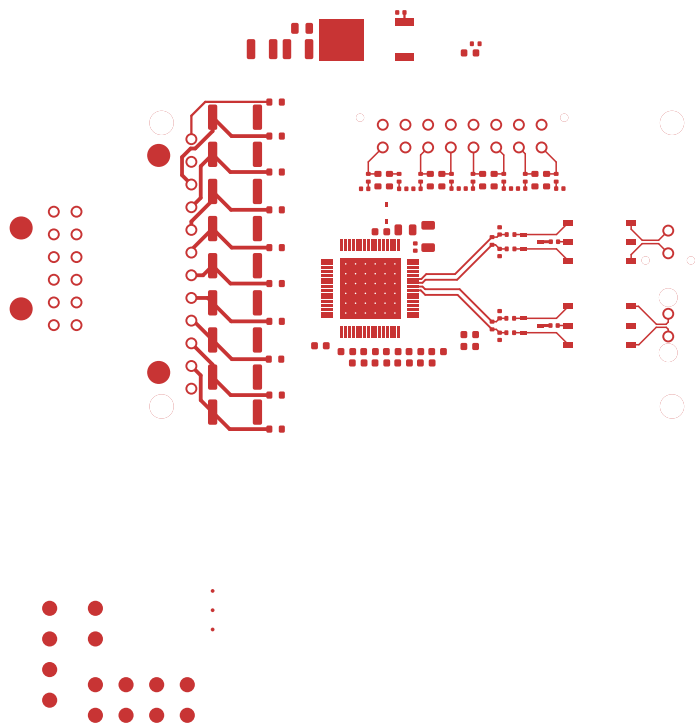


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			Blue Devil Motor Sports		PRELIMINARY		a06d4b7		
			Board Name:			Project Name:			
			BMS_SLAVE			BMS Slave			
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Bottom Test Points (Scale 1:1)		bms_slave.kicad_pcb		Aidan Brzezinski		2024-04-13		+ (Unreleased)	
Sheet Path:				Reviewer:		Size:		Sheet:	
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L1 (Sig, PWR) (Scale 1:1)



Comments:

Company:

Blue Devil Motor Sports

Variant:

PRELIMINARY

Git Hash:

a06d4b7

Board Name:

BMS_SLAVE

Project Name:

BMS Slave

Sheet Title:

L1 (Sig, PWR) (Scale 1:1)

File Name:

bms_slave.kicad_pcb

Designer:

Aidan Brzezinski

Date:

2024-04-13

Revision:

+ (Unreleased)

Sheet Path:

Reviewer:

Size:

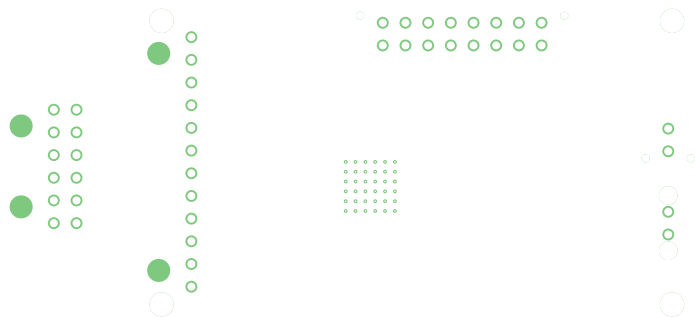
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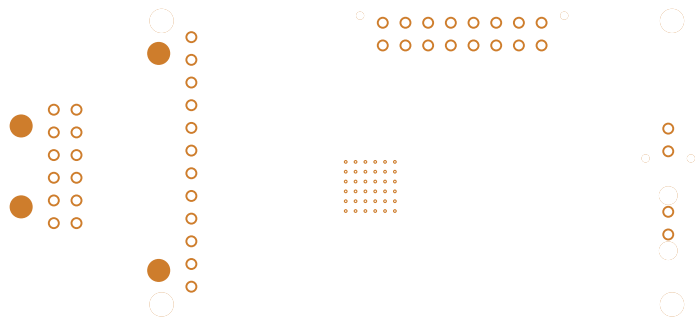
L2 (GND) (Scale 1:1)



	Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: a06d4b7
	Sheet Title: L2 (GND) (Scale 1:1)	Board Name: BMS_SLAVE		Project Name: BMS Slave	
	Sheet Path:	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
			Reviewer:	Size: A4	Sheet: 8 of 10

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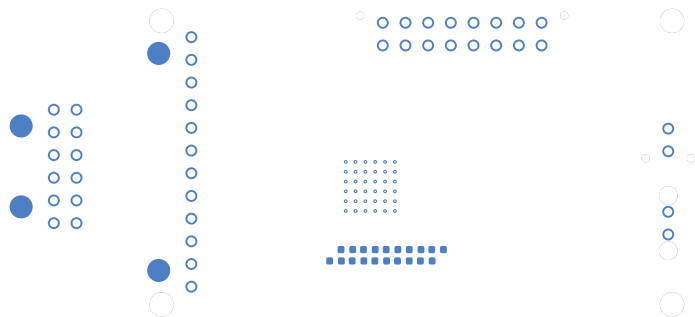
L3 (Sig, PWR) (Scale 1:1)



	Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: a06d4b7
	Sheet Title: L3 (Sig, PWR) (Scale 1:1)	Board Name: BMS_SLAVE	Project Name: BMS Slave		
	Sheet Path:	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
			Reviewer:	Size: A4	Sheet: 9 of 10

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L4 (Sig, PWR) (Scale 1:1)



	Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: a06d4b7
	Sheet Title: L4 (Sig, PWR) (Scale 1:1)	Board Name: BMS_SLAVE	Project Name: BMS Slave		
	Sheet Path:	File Name: bms_slave.kicad_pcb	Designer: Aidan Brzezinski	Date: 2024-04-13	Revision: + (Unreleased)
			Reviewer:	Size: A4	Sheet: 10 of 10